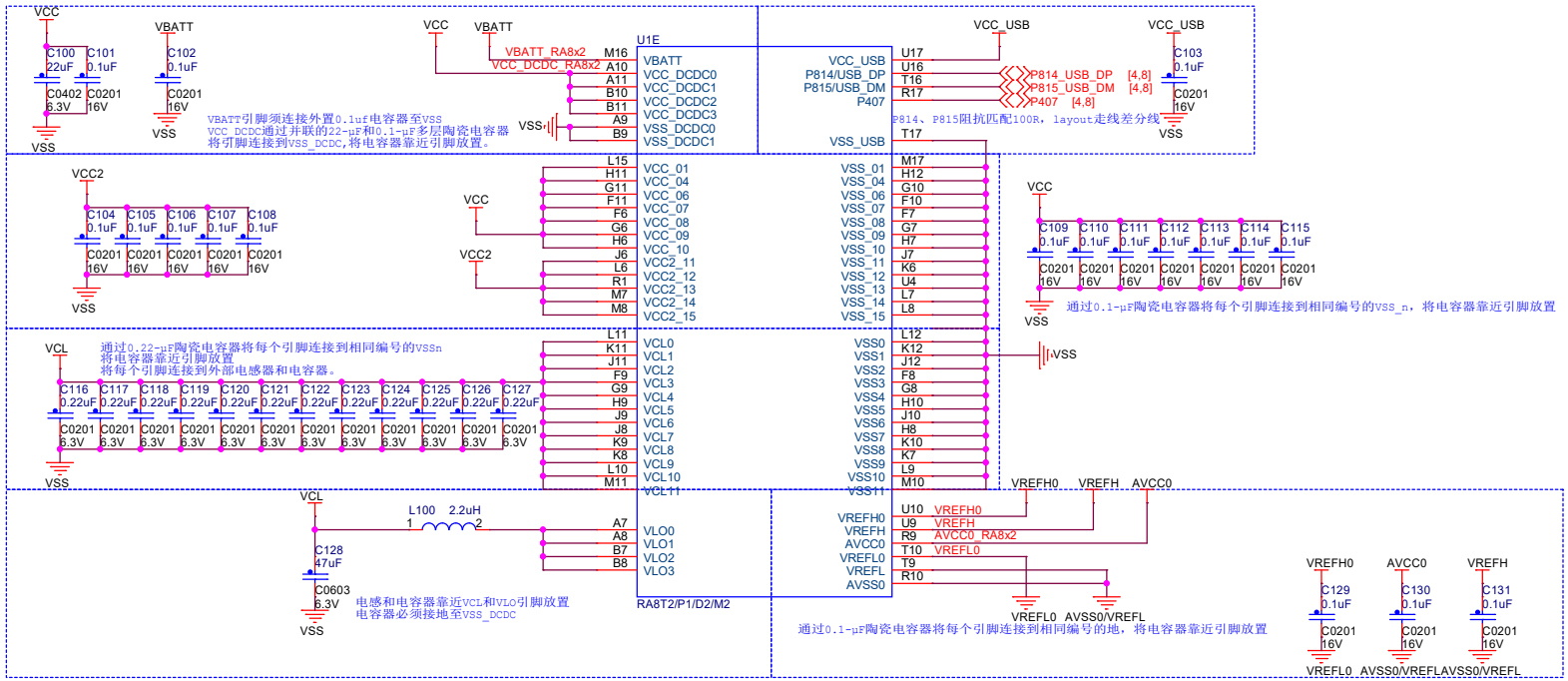
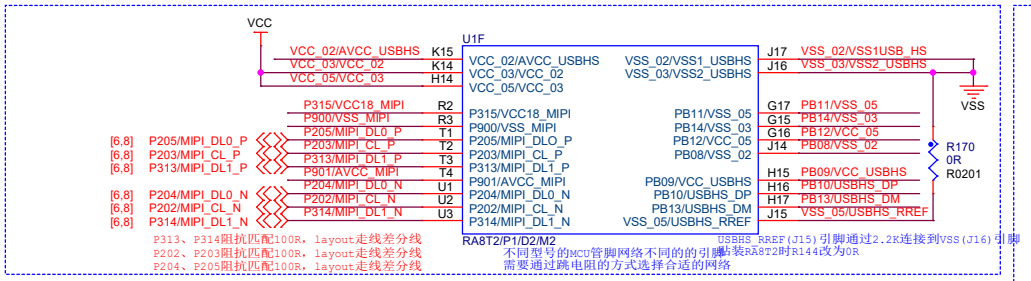


Page1: MCU 电源
Page2: MCU I/O
Page3: Jlink OB
Page4: USB HS
Page5: SDRAM
Page6: STORAGE MEMORY
Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB

核心板缺省功能阻容设置

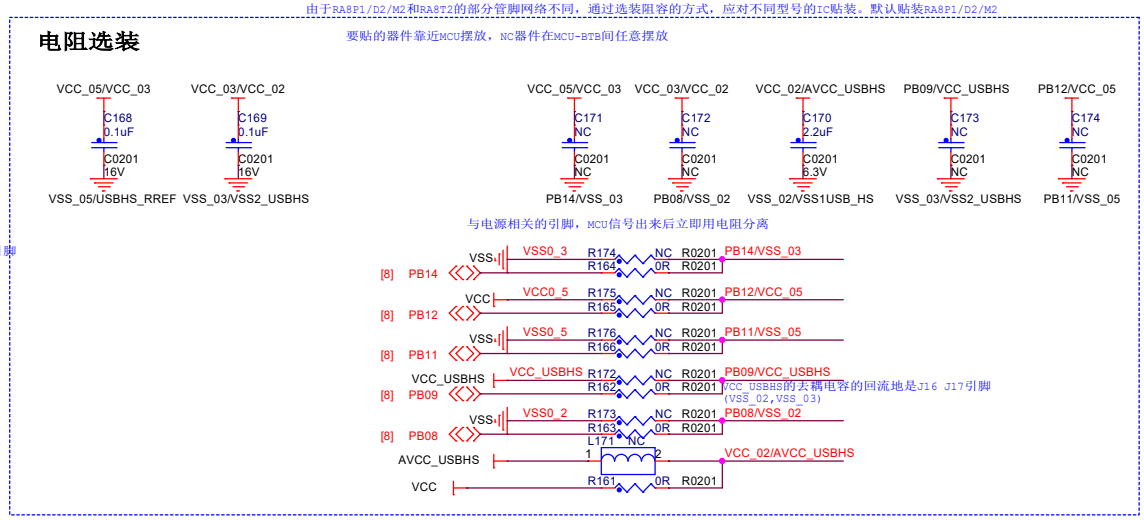
核心板芯片	R7KA8T2L	R7KA8P1K
C168-169	0.1uF	NC
C170	2.2uF	2.2uF
C171-C174	NC	0.1uF
C191-C192	NC	0.1UF
R145-R150	NC	NC
R151-R168	0R	NC
R170	0R	2.20K 1%
L171	NC	120R 100MHZ
R172-R180	NC	0R
R191-R193	NC	0R

由于RA8P1/D2/M2和RA8T2的部分管脚网络不同，通过选装阻容的方式，应对不同型号的IC贴装。默认贴装RA8P1/D2/M2



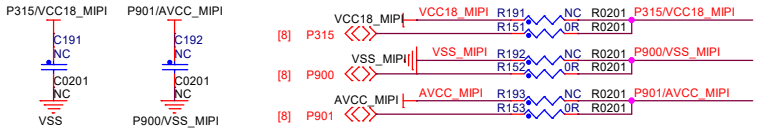
电阻选装

要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放

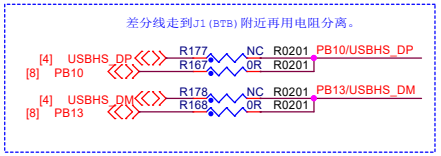


MIPI部分

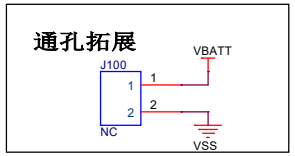
要贴的器件靠近MCU摆放，NC器件在MCU-BTB间任意摆放



差分线走到J1(BTB)附近再用电阻分离。



通孔拓展



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Title	野火瑞萨RA8T2-BTB核心板(PMIC)_原理图-供WS贴装用	
Size A3	Document Number MCU 电源	Rev V0.1
Date: Wednesday, January 07, 2026	Sheet 1 of 11	

Page1: MCU 电源
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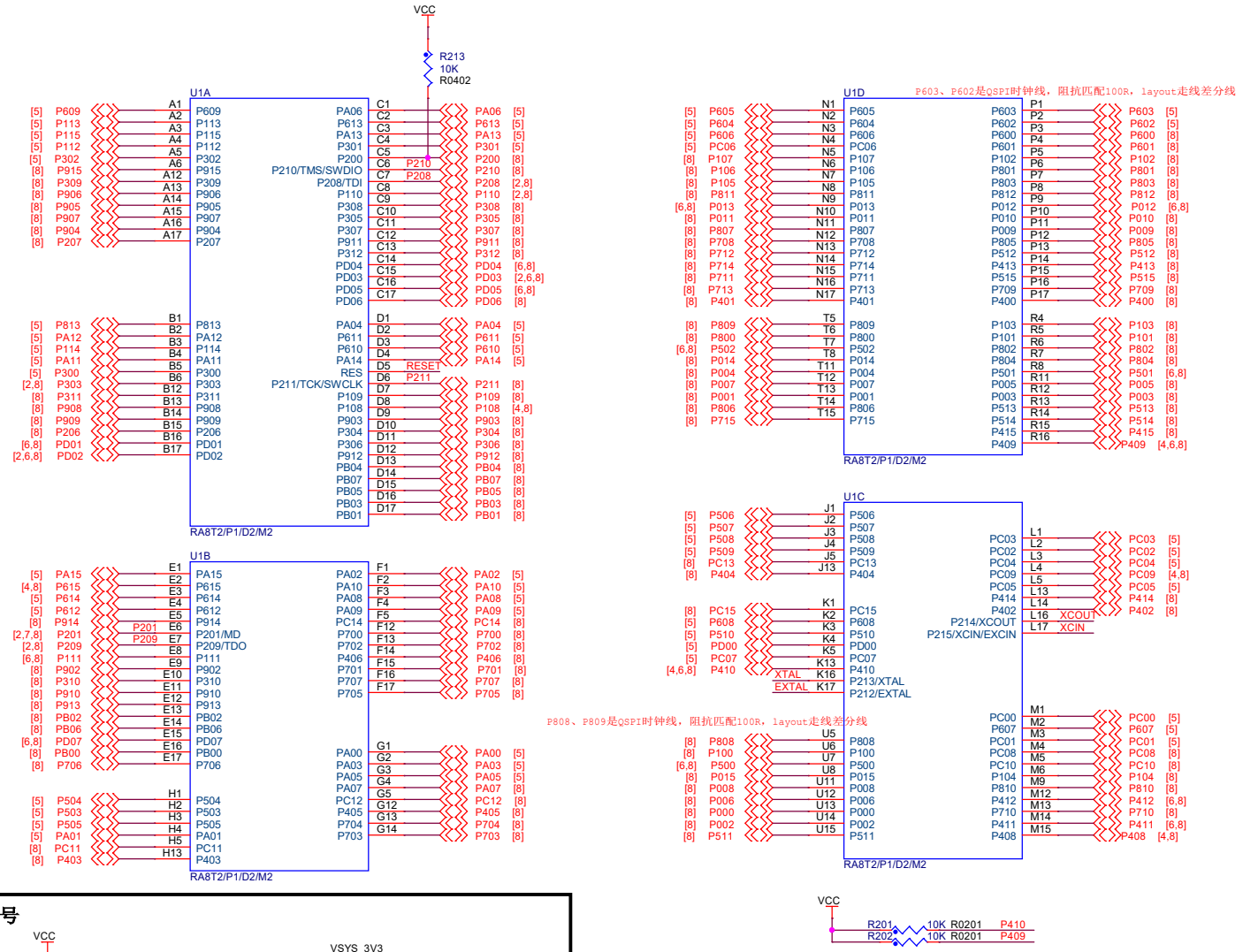
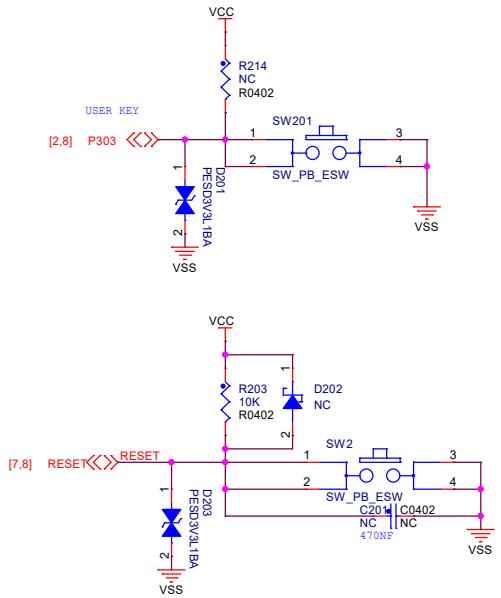
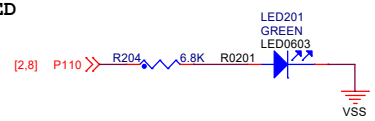


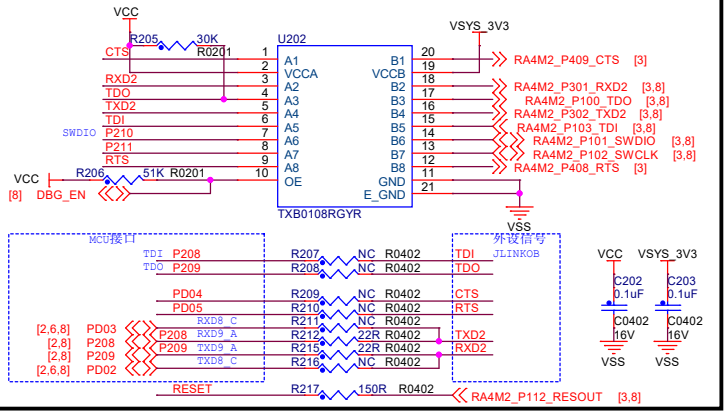
用户按键



LED

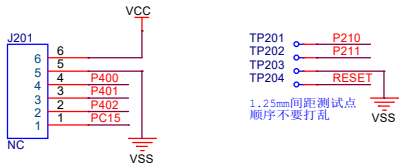


接口信号



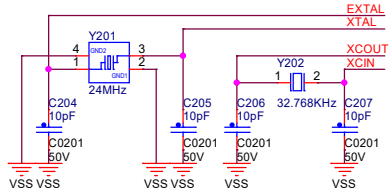
通孔拓展

VIN有大电流需求，请用粗线或铜皮连到VIN网络。
J201，J202之间的间距请保证2.54mm的倍数



如果使用USB HS，主晶振只能选12/20/24/48MHz

晶振



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Title	野火瑞萨RA8T2-BTB核心板(PMIC)_原理图-供WS贴装用	
Size A3	Document Number MCU IO	Rev V0.1
Date: Monday, January 05, 2026	Sheet 2 of 11	

Jlink OB

注意：R7FA4M2只能12MHz晶振

Page1: MCU 电源
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Page7: 电源LDO/DCDC/MIPI/FLASH
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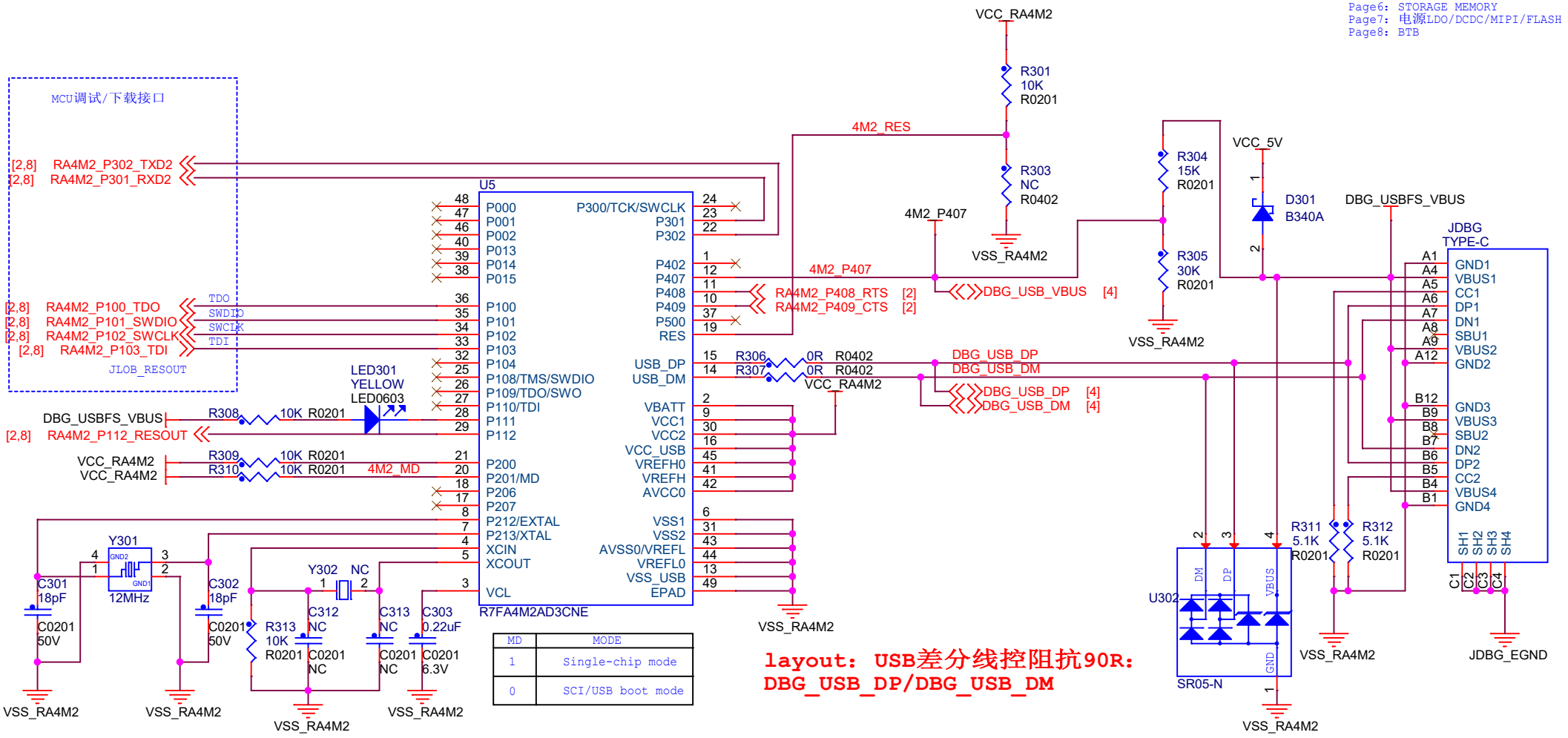
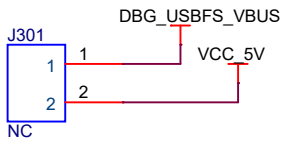
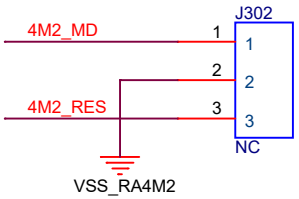


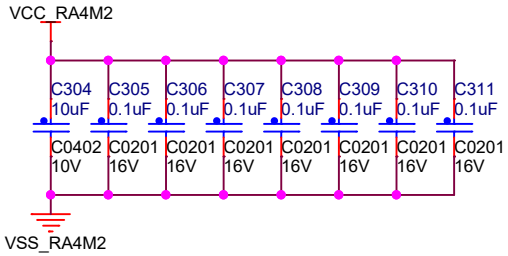
MD	MODE
1	Single-chip mode
0	SCI/USB boot mode

测试点 1.25间距/预留



debug隔离工作时，用作短路二极管运行





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Size A4	Document Number Jlink OB	Rev V0.1
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USB HS

如果使用USB HS，主晶振只能选12/20/24/48MHz

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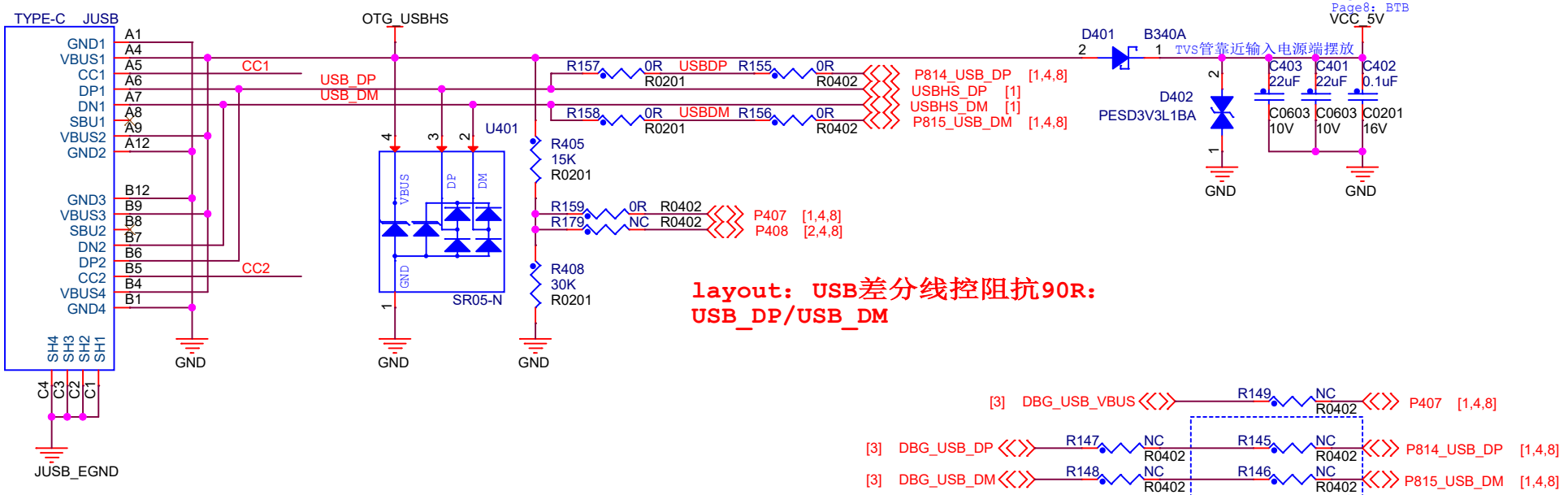
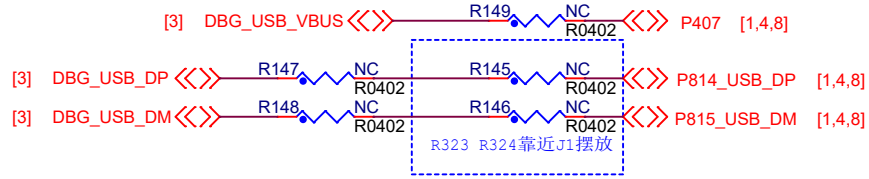
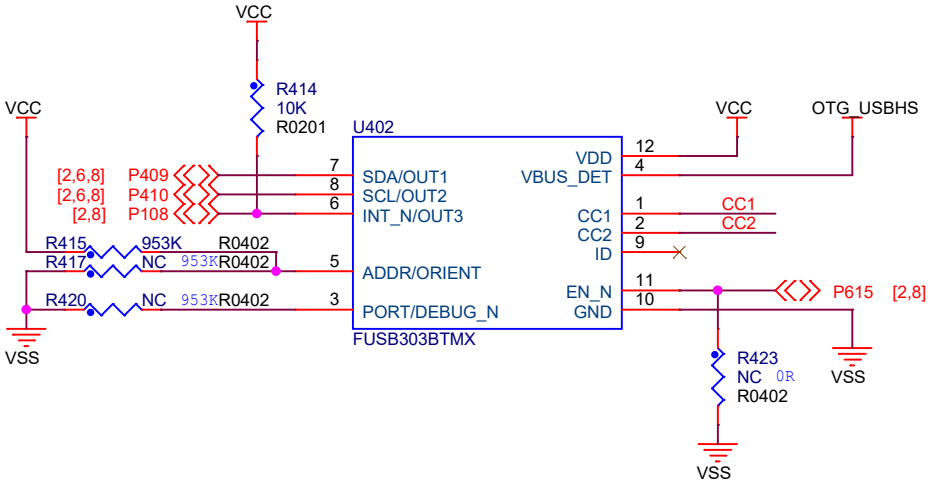
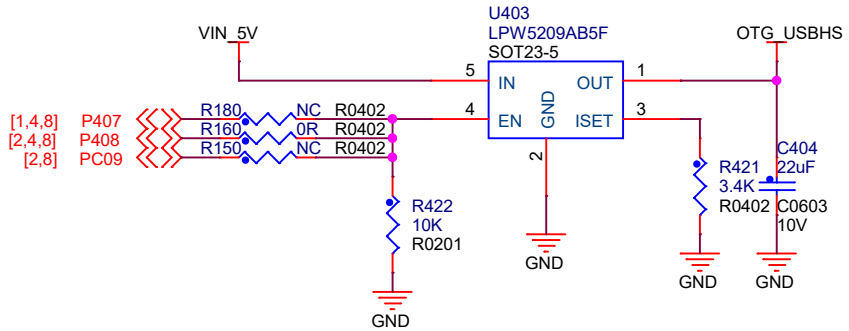


layout: USB差分线控阻抗90R:
USB_DP/USB_DM







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Title 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图-供WS贴装用		
Size A4	Document Number USB HS	Rev V0.1
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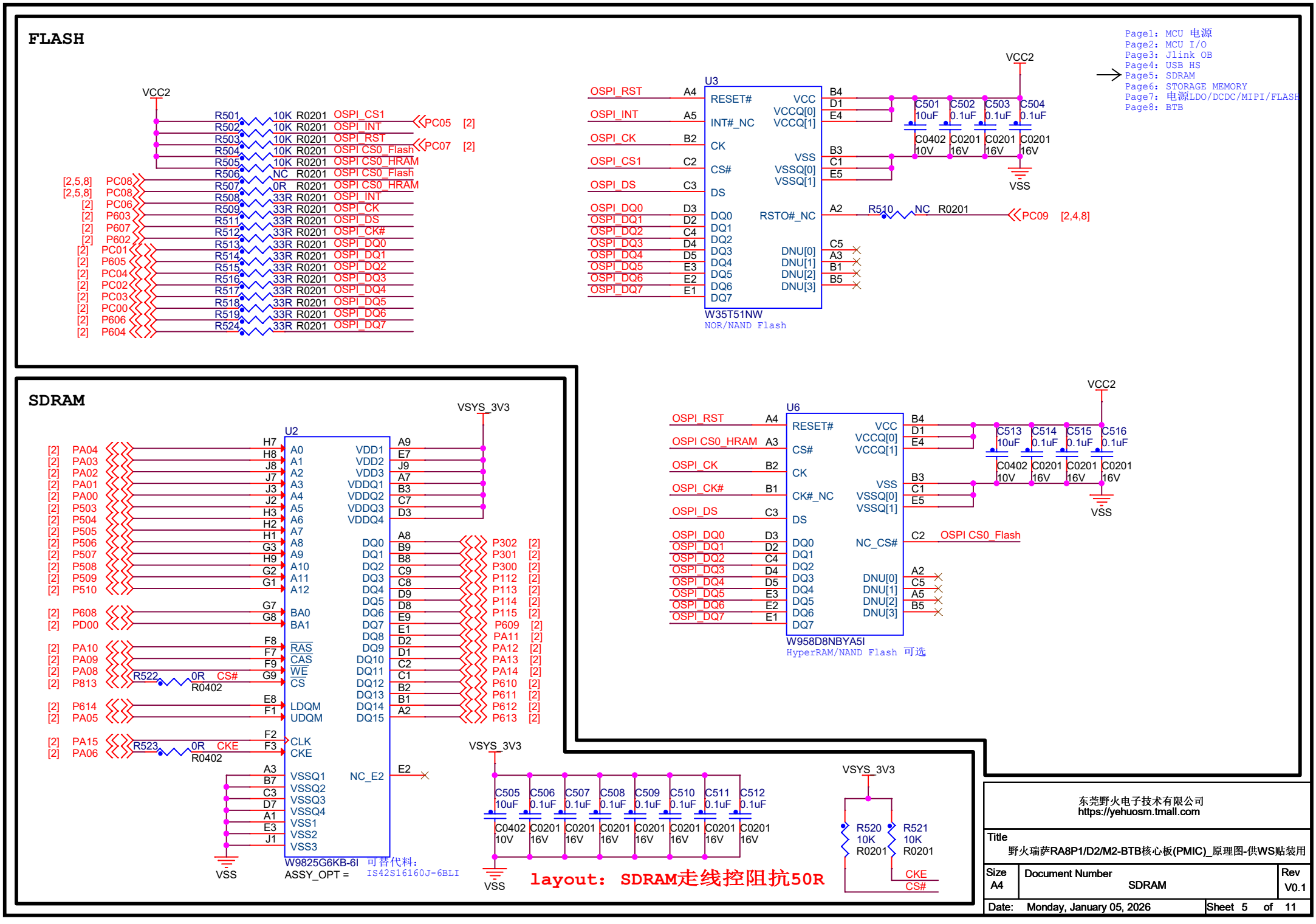
FLASH
VCC2
R501 10K R0201 OSPI_CS1
R502 10K R0201 OSPI_INT
R503 10K R0201 OSPI_RST
R504 10K R0201 OSPI CS0_Flash
R505 10K R0201 OSPI CS0_HRAM
R506 NC R0201 OSPI CS0_Flash
R507 0R R0201 OSPI CS0_HRAM
R508 33R R0201 OSPI_INT
R509 33R R0201 OSPI_CK
R511 33R R0201 OSPI_DS
R512 33R R0201 OSPI_CK#
R513 33R R0201 OSPI_DQ0
R514 33R R0201 OSPI_DQ1
R515 33R R0201 OSPI_DQ2
R516 33R R0201 OSPI_DQ3
R517 33R R0201 OSPI_DQ4
R518 33R R0201 OSPI_DQ5
R519 33R R0201 OSPI_DQ6
R524 33R R0201 OSPI_DQ7
PC05 [2]
PC07 [2]
[2,5,8] PC08
[2,5,8] PC08
[2] PC06
[2] P603
[2] P607
[2] P602
[2] PC01
[2] P605
[2] PC04
[2] PC02
[2] PC03
[2] PC00
[2] P606
[2] P604
U3
OSPI_RST A4
OSPI_INT A5
OSPI_CK B2
OSPI_CS1 C2
OSPI_DS C3
OSPI_DQ0 D3
OSPI_DQ1 D2
OSPI_DQ2 C4
OSPI_DQ3 D4
OSPI_DQ4 D5
OSPI_DQ5 E3
OSPI_DQ6 E2
OSPI_DQ7 E1
RESET# INT#_NC CK CS# DS DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7
VCC VCCQ[0] VCCQ[1] VSS VSSQ[0] VSSQ[1] RSTO#_NC DNU[0] DNU[1] DNU[2] DNU[3]
B4 D1 E4 B3 C1 E5 A2 C5 A3 B1 B5
C501 10uF C502 0.1uF C503 0.1uF C504 0.1uF
C0402 10V C0201 16V C0201 16V C0201 16V
VSS
R510 NC R0201
PC09 [2,4,8]
W35T51NW
NOR/NAND Flash
Page1: MCU 电源
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Page7: 电源LDO/DCDC/MIPI/FLASH
Page8: BTB
SDRAM
VSYS_3V3
U2
[2] PA04 H7 A0
[2] PA03 H8 A1
[2] PA02 J8 A2
[2] PA01 J7 A3
[2] PA00 J3 A4
[2] P503 J2 A5
[2] P504 H3 A6
[2] P505 H2 A7
[2] P506 H1 A8
[2] P507 G3 A9
[2] P508 H9 A10
[2] P509 G2 A11
[2] P510 G1 A12
[2] P608 G7 BA0
[2] PD00 G8 BA1
[2] PA10 F8 RAS
[2] PA09 F7 CAS
[2] PA08 F9 WE
[2] P813 G9 CS
R522 0R CS# R0402
[2] P614 E8 LDQM
[2] PA05 F1 UDQM
[2] PA15 F2 CLK
[2] PA06 F3 CKE
R523 0R CKE R0402
A3 B7 C3 D7 A1 E3 J1
VSS1 VSS2 VSS3
VDD1 VDD2 VDD3 VDDQ1 VDDQ2 VDDQ3 VDDQ4
A9 E7 J9 A7 B3 C7 D3
DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7 DQ8 DQ9 DQ10 DQ11 DQ12 DQ13 DQ14 DQ15
A8 B9 B8 C9 C8 D9 D8 E9 E1 D2 D1 C2 C1 B2 B1 A2
P302 [2] P301 [2] P300 [2] P112 [2] P113 [2] P114 [2] P115 [2] P609 [2] PA11 [2] PA12 [2] PA13 [2] PA14 [2] P610 [2] P611 [2] P612 [2] P613 [2]
NC_E2 E2
VSS
W9825G6KB-6I 可替代料:
ASSY_OPT = IS42S16160J-6BLI
U6
OSPI_RST A4
OSPI CS0_HRAM A3
OSPI_CK B2
OSPI_CK# B1
OSPI_DS C3
OSPI_DQ0 D3
OSPI_DQ1 D2
OSPI_DQ2 C4
OSPI_DQ3 D4
OSPI_DQ4 D5
OSPI_DQ5 E3
OSPI_DQ6 E2
OSPI_DQ7 E1
RESET# CS# CK CK#_NC DS DQ0 DQ1 DQ2 DQ3 DQ4 DQ5 DQ6 DQ7
VCC VCCQ[0] VCCQ[1] VSS VSSQ[0] VSSQ[1] NC_CS# DNU[0] DNU[1] DNU[2] DNU[3]
B4 D1 E4 B3 C1 E5 C2 A2 C5 A5 B5
OSPI CS0_Flash
C513 10uF C514 0.1uF C515 0.1uF C516 0.1uF
C0402 10V C0201 16V C0201 16V C0201 16V
W958D8NBYA5I
HyperRAM/NAND Flash 可选
C505 10uF C506 0.1uF C507 0.1uF C508 0.1uF C509 0.1uF C510 0.1uF C511 0.1uF C512 0.1uF
C0402 10V C0201 16V C0201 16V C0201 16V C0201 16V C0201 16V C0201 16V C0201 16V
R520 10K R0201 R521 10K R0201
CKE
CS#
layout: SDRAM走线控阻抗50R
东莞野火电子技术有限公司
https://yehuosm.tmall.com
Title
野火瑞萨RA8P1/D2/M2-BTB核心板(PMIC)_原理图-供WS贴装用
Size A4
Document Number SDRAM
Rev V0.1
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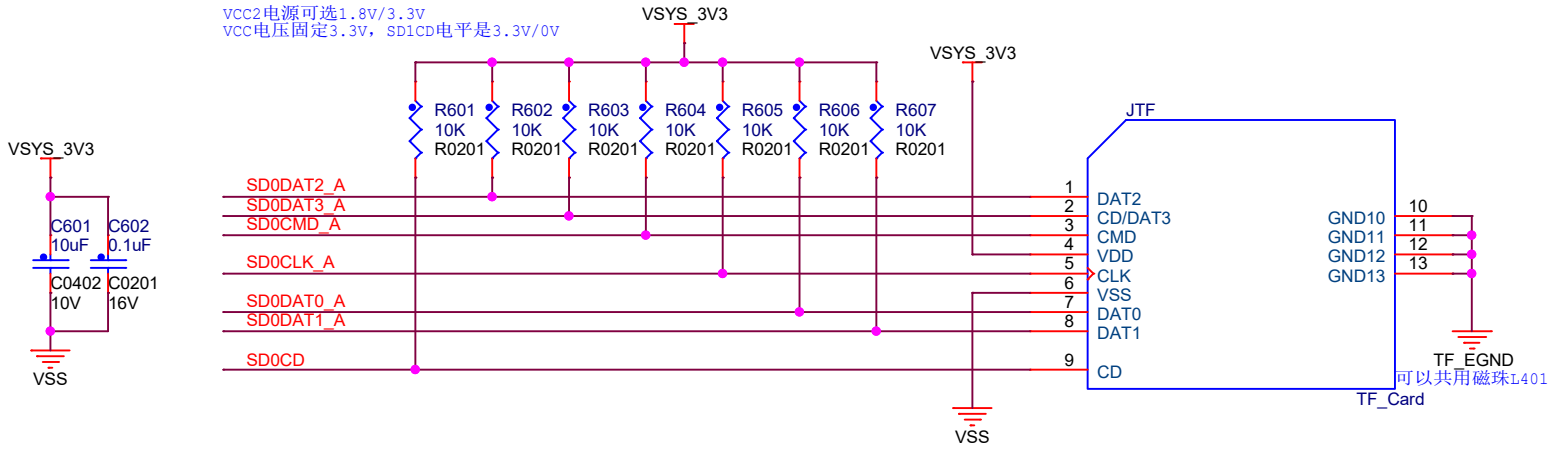
SD卡



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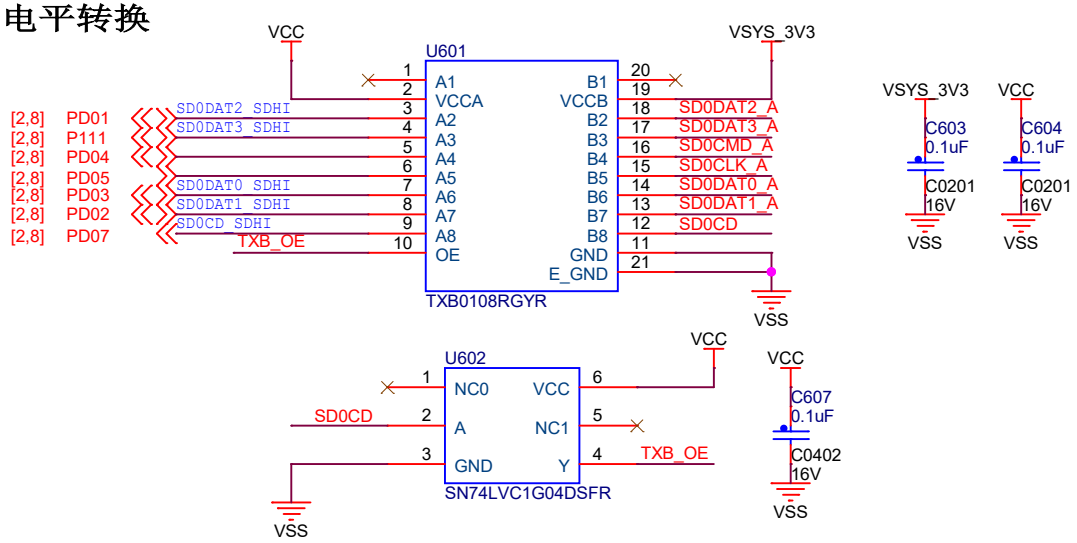
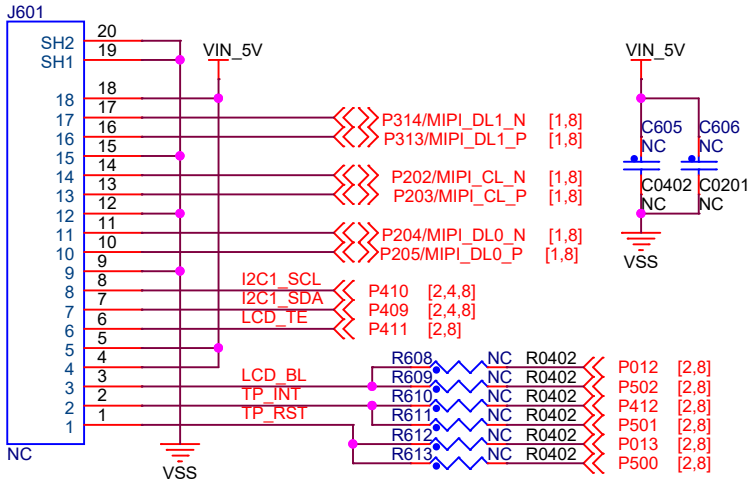
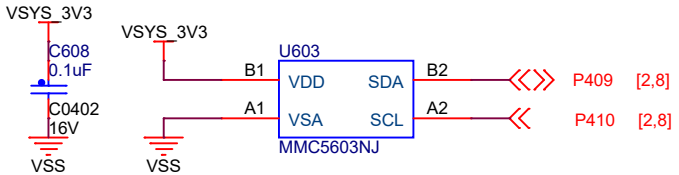
SD卡电平转换



MIPI屏



地磁传感器



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Title 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图-供WS贴装用		
Size A4	Document Number STORAGE MEMORY	Rev V0.1
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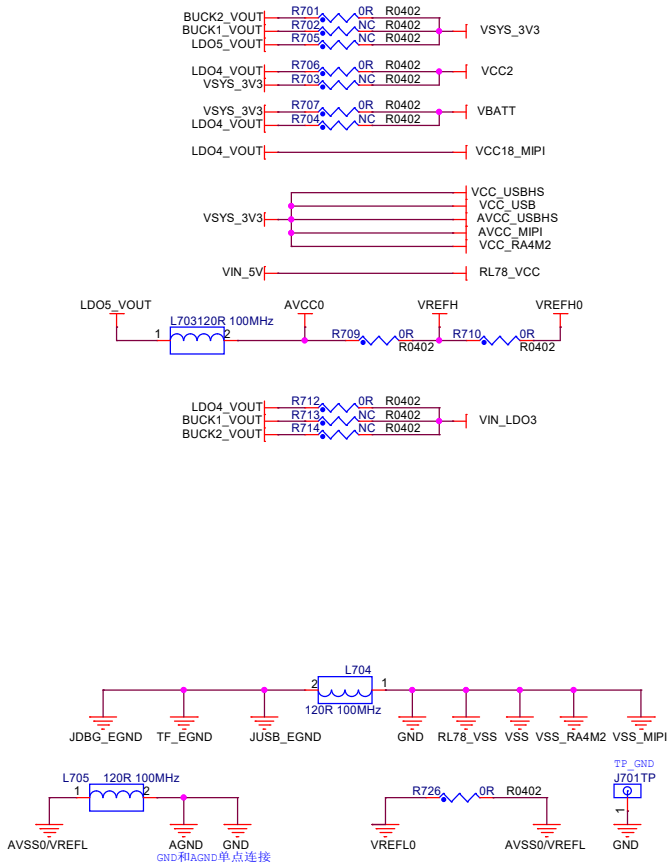
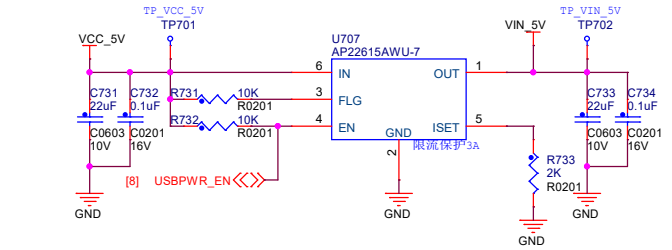
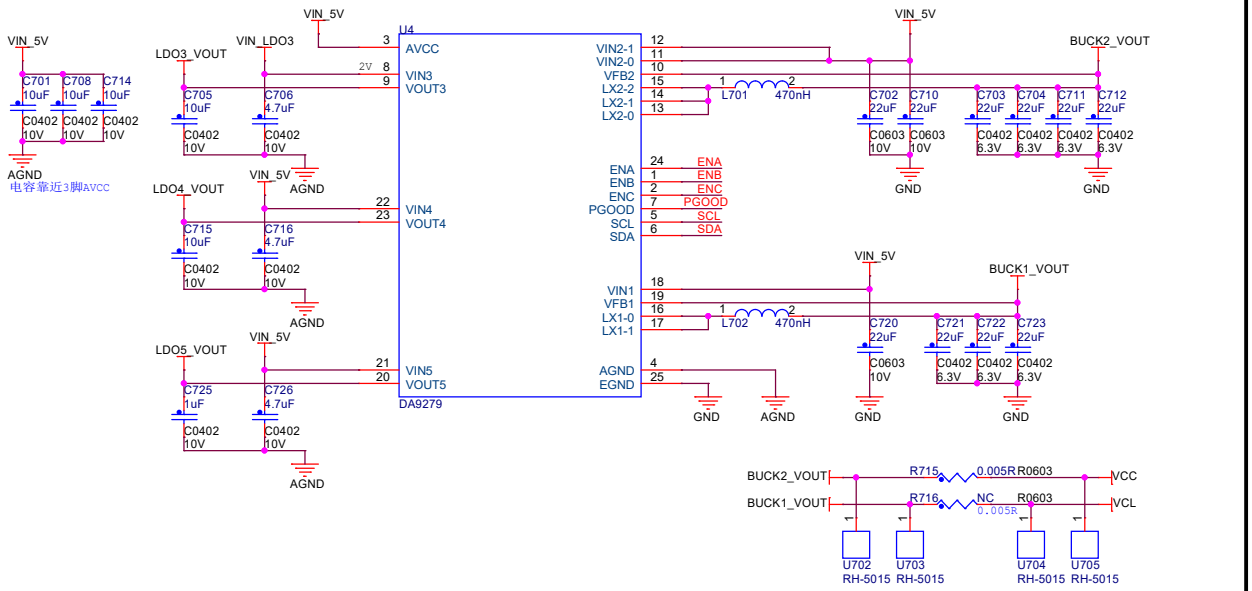
电源分割



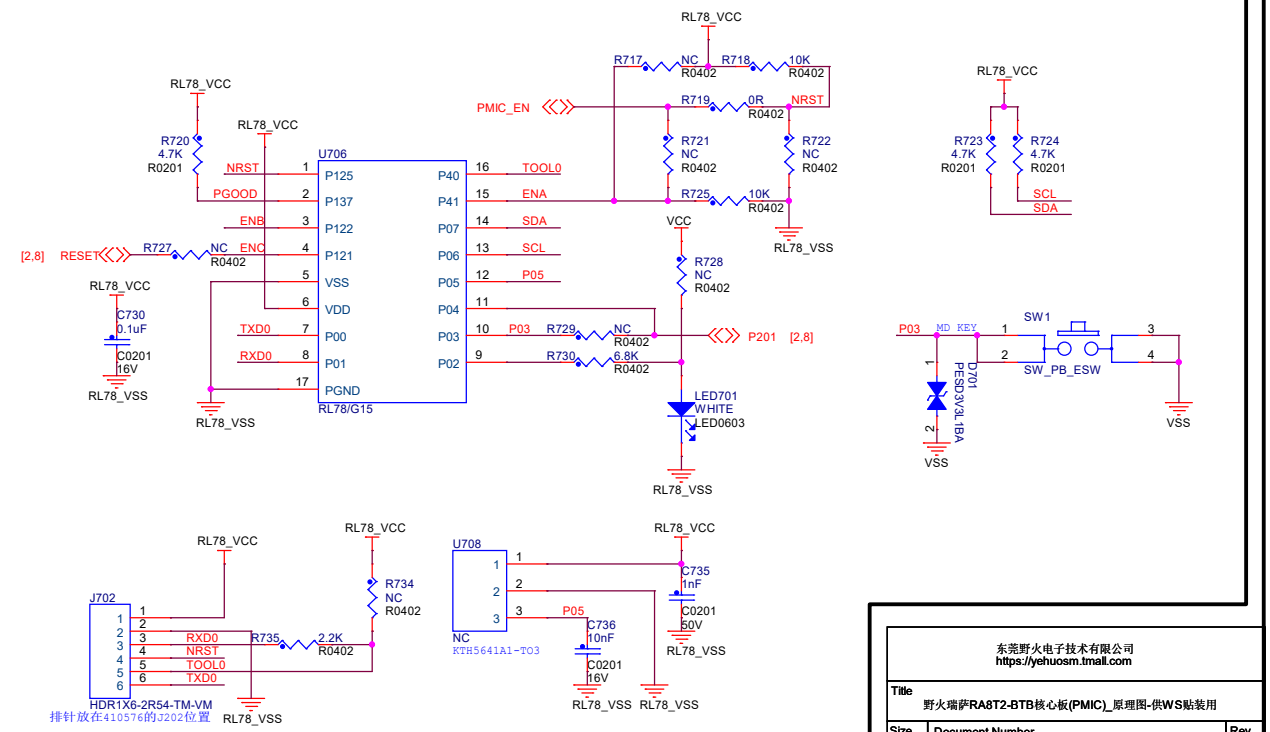
电源保护



PMIC



MCU



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Title 野火瑞萨RA8T2-BTB核心板(PMIC)_原理图-供WS贴装用		
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BTB座子

Page1: MCU 电源
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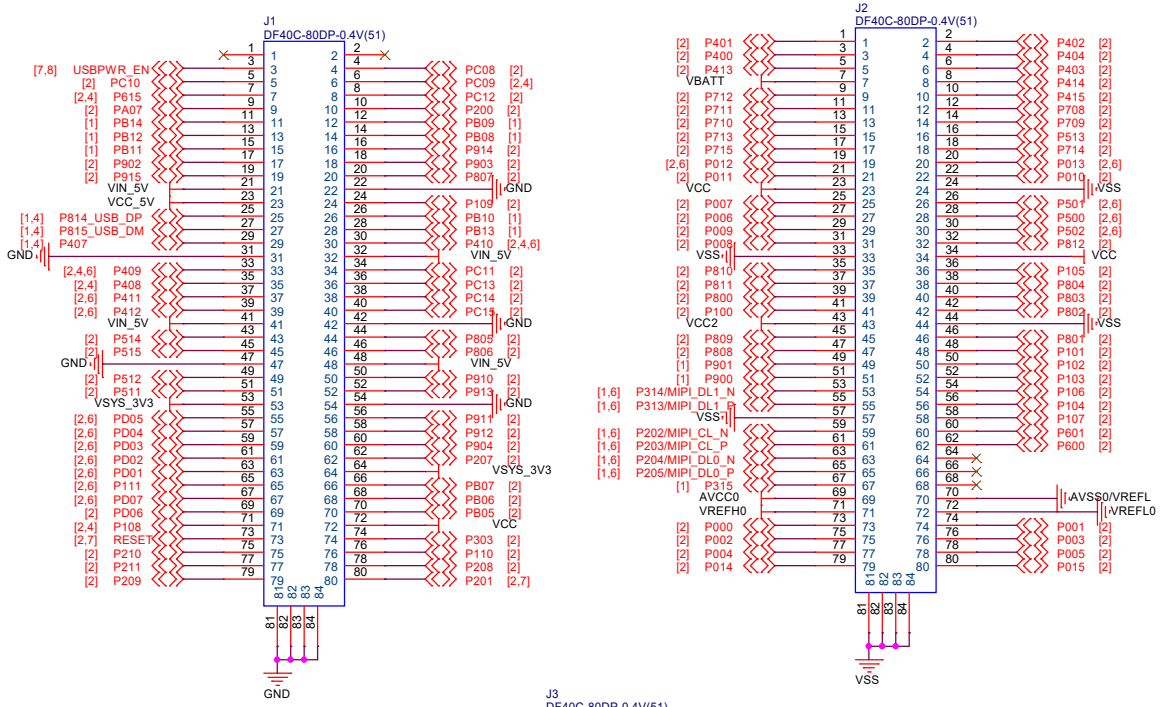
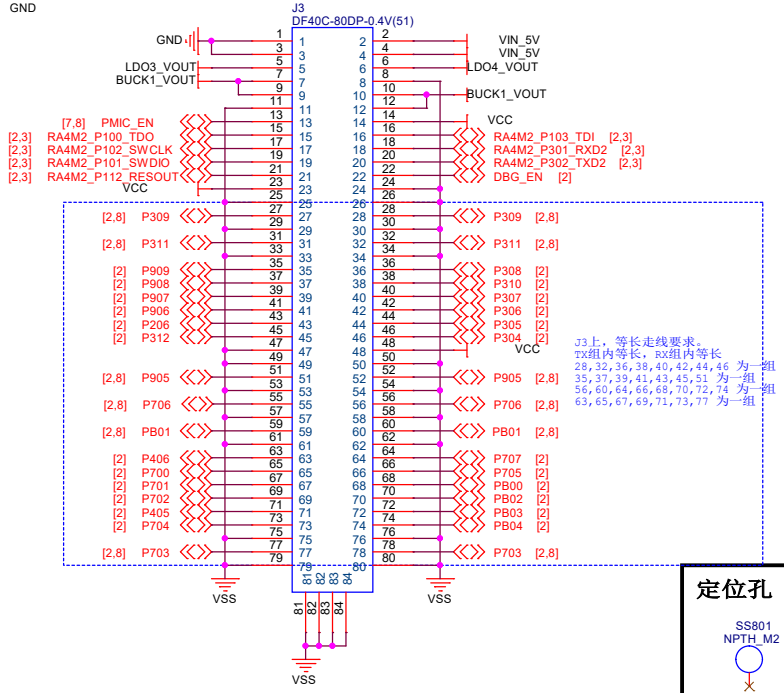
Page8: BTB





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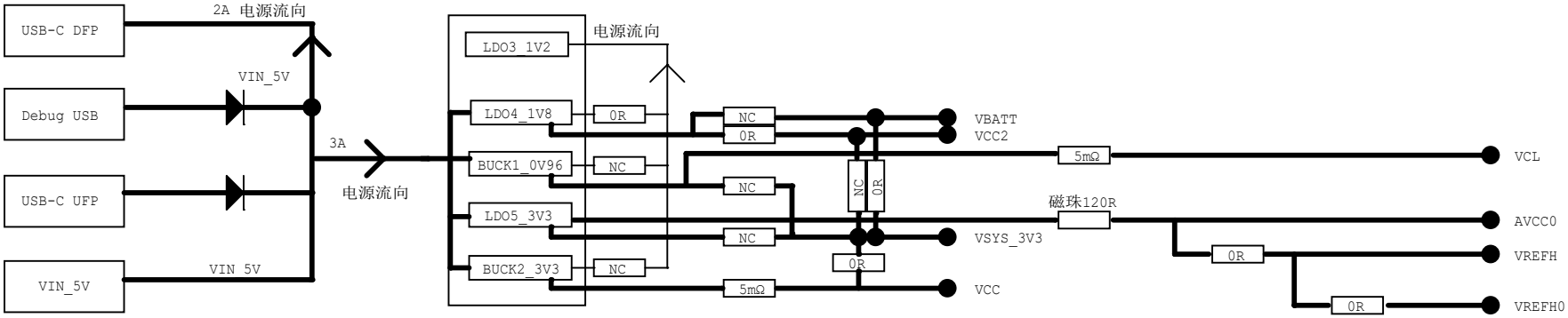
电源拓扑图



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历史版本

版本号	日期	设计	描述
V0.1	2025-05-19	LPC	初始版本

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